



# Amine Ezzine

Double Master's Graduate In Aerospace and Physics Engineering

## CONTACT

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## Core Skills

- **Electromagnetics & Electrotechnics**
- **Power Transport Systems**
- **Numerical Simulation & FEM**
- **Signal Processing & Automation**
- **Thermodynamics & Fluid Dynamics**

## Technical Skills

- **Languages:** Python, MATLAB, C, Java, SQL
- **Simulation & FEA:** COMSOL, Xflow, Catia, Fusion 360
- **Documentation:** LaTeX, MS office
- **OS:** Windows, Linux

## Languages

- **English:** Fluent
- **French:** Fluent
- **Arabic:** Native
- **Swedish:** Beginner
- **Japanese / Sign Language:** Beginner

## Honours

- **Academic Excellence**
- **Scholarship:** ISAE-SUPAERO Foundation
- **Tunisian Embassy Scholarship** Tunisian Government

## Interests

- **Our Giant Leap Podcast:**  
Experience as podcast host for the diversity and gender equality project group( Space generation advisory council) podcast's
- **Debate:** President of ISAE-SUPAERO Debate Club

## Summary

My background combines Aerospace Engineering (ISAE-SUPAERO) and Engineering Physics (KTH), with a focus on numerical simulation, electromagnetism, thermodynamics, and complex physical systems. Through industrial and academic projects, I have developed simulation tools, analyzed experimental data, and translated physical phenomena into engineering solutions. My interests lie in applying numerical simulation and physical modelling to the development and validation of innovative technologies

## Professional Experience

### Research Intern, DEOS Laboratory ISAE-SUPAERO

Toulouse, France (2024, 6 months)

- *Developed analytical, finite-element (COMSOL), and MATLAB models for electromagnetic wave propagation in complex media.*
- *Investigated physics-based optimization methods to reduce propagation energy losses*
- *Studied the effect of frequency, wavefront shape, and medium properties on signal absorption.*
- *Demonstrated up to 10% reduction in energy loss for optimized wavefront configurations.*

### Detectors calibration trainee, Sysnav

Vernon, France (2023, 3 months)

- *Automated thermal and impact test-data analysis pipelines for electromagnetic localization devices using Python*
- *Developed tools for raw-data cleaning, metric extraction, and defect detection*
- *Improved efficiency of detector validation workflows and reduced manual analysis workload*
- *Contributed to the foundation of future AI-assisted diagnostic and calibration tools*

### Master's Thesis, Stockholm University

Stockholm, Sweden, (2025, 6 months)

- *Developed numerical models to simulate wave propagation and field interactions in complex physical environments.*
- *Conducted quantitative parameter studies and model validation using experimental and literature-based constraints.*
- *Performed parameter estimation using experimental constraints to identify viable theory regimes.*
- *Produced technical documentation and presented results to supervisors and collaborators to support project decision-making.*

## EDUCATION

### Aerospace Engineering Degree

ISAE-SUPAERO, Toulouse, France, (2023 - 2026)

- *Advanced coursework in signal processing, fluid dynamics, PDEs, aerodynamics, aeroplane control, and finite elements.*
- *Laboratory work in thermodynamics, circuit design, and thermo analysis of propulsion systems using CFD simulations.*
- *Data analysis and computational methods using Python, MATLAB, and SQL.*

### Master of Science In Engineering Physics

KTH, Stockholm, Sweden, (2024-2026)

- *Advanced quantum mechanics, relativity, and theoretical particle physics*
- *Experimental particle physics using ROOT framework for data analysis*
- *Strong focus on theoretical modelling and mathematical physics*

### Bachelor in Mathematics, Physics, and Engineering

ISAE-SUPAERO, Toulouse, France

- *Coursework in Power transport system, electric circuit design, signal processing, linear algebra, numerical methods.*
- *Engineering labs in circuits, thermodynamics, and system design and power transport systems.*